Lab Manual - LAB 01

Semester V – Fall

Group A and B

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Bachelor of Science in Software Engineering

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LAB 1: Computer Networking Foundations

Table of Contents

Computer Networks....3
Why Do We Need Computer Networks?....4
How Does Network Communication Work?....5
A Real-World Example: Opening a Website....7
Basic Networking Terminology....7
Wired and Wireless Networks....13
Network Types Based on Geographic Area....15
Internet....17
Intranet....17
    Internet vs Intranet....17
Network Components and Devices....18
MAC Address and IP Address....22
Network Ports....23
Basic Networking Protocols....23
Why Do We Need Networking Models?....24
OSI Model....25
Easy Way to Understand the OSI Model....27
TCP/IP Model....28
OSI vs TCP/IP....30

Introduction to Computer Networks

Learning Objectives

By the end of this week, students should be able to:

1. Explain what a computer network is and why networks are important.
2. Describe basic networking concepts and terminology.
3. Differentiate between wired/wireless networks, LAN, MAN, WAN, Internet, and Intranet.
4. Identify common network components and explain their basic functions.
5. Explain the purpose of MAC addresses, IP addresses, ports, and basic networking protocols.
6. Describe the OSI Reference Model and TCP/IP Model at a basic conceptual level.

Computer Networks

A **computer network** is a collection of two or more connected devices that can communicate with each other and exchange data or share resources.

A network can contain different types of devices, such as:

- Computers
- Laptops
- Smartphones
- Servers
- Printers
- Switches
- Routers
- Wireless access points

Suppose 20 computers are connected to a switch. The computers can communicate with each other and may also access a shared printer or the Internet. This collection of connected devices forms a **computer network**.

In simple words, a computer network allows devices to communicate and share resources.

Why Do We Need Computer Networks?

Networks are an essential part of modern computing because individual computers cannot efficiently operate in isolation.

1. Communication

Networks allow users and devices to communicate.
Examples:

- Sending emails
- Video conferencing
- Instant messaging
- Voice calls
- Sharing information

2. Resource Sharing

Multiple users can share resources through a network.
Examples:

- Network printers
- Storage
- Internet connection
- Applications
- Databases

For example, instead of connecting a printer separately to every computer in a laboratory, the printer can be connected to the network and shared by all computers.

3. Data Sharing

Users can transfer information between computers.
For example: **Computer A → Network → Computer B**

Computer A might send:

- A document
- An image

- A video
- A database record

4. Centralized Services

Organizations can keep important services on centralized servers.
For example:

Students → University Network → University Server

The server might provide:

- Student portal
- Learning management system
- Database
- File storage
- Authentication

5. Internet Access

A network allows devices to access services outside the local environment.
For example:

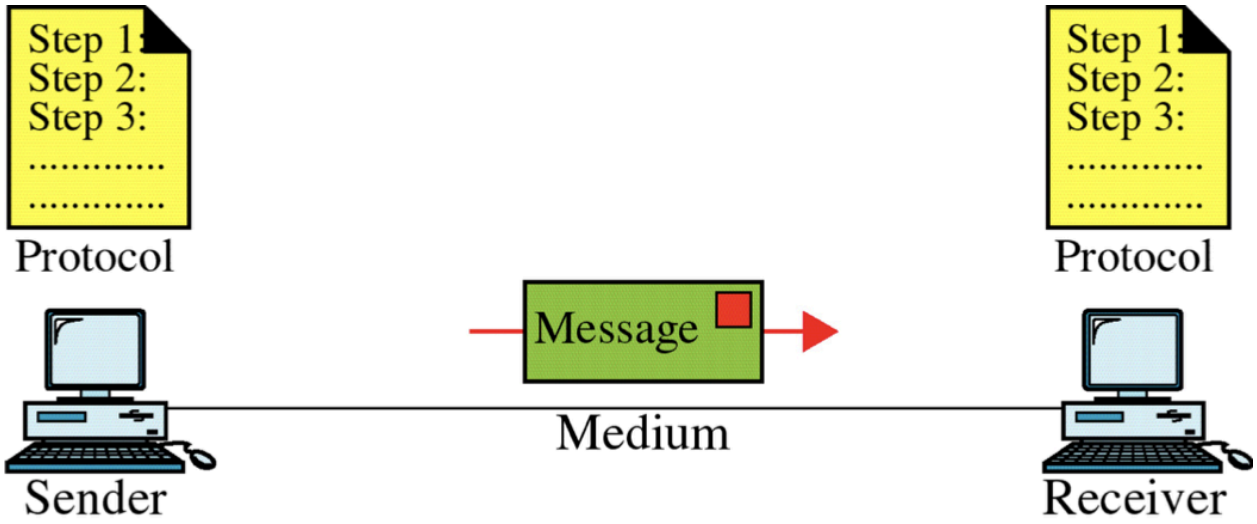
Laptop → Wi-Fi Router → ISP → Internet

This is how a laptop can access websites hosted on servers around the world.

How Does Network Communication Work?

Before learning complicated networking concepts, it is important to understand the basic communication process. A data communication system is made up of 5 components:

1. Message
2. Sender
3. Receiver
4. Medium
5. Protocol



1. **Message**

   The message is the information (data) to be communicated. A message is transmitted from sender to receiver. It can be

   - data file
   - an audio or video file
   - or any text message

   "A message also keeps information like sender and destination addresses."

2. **Sender**

   The **sender** is the device that generates and sends information. Examples:

   - Computer sending a file
   - Smartphone sending a message
   - Browser requesting a webpage

3. **Receiver**

   The **receiver** is the device that receives the transmitted information. Examples:

   - Computer receiving a file
   - Smartphone receiving a message
   - Web server receiving a request

4. **Communication Medium**

   The **communication medium** is the path through which data travels. It can be wired and wireless.

   - Wired
     - Data travels through a physical medium such as a network cable.
     - Example: **PC → Ethernet Cable → Switch**
   - Wireless
     - Data travels using wireless signals.
     - Example: **Laptop → Wi-Fi → Access Point**

5. **Protocol**

   Protocols are also called language of communication. A protocol is a set of rules that governs data communication.

   - Both sender & receiver must have the same protocol for communication.
   - It represents an agreement between the communicating devices.
   - Without a protocol, two devices may be connected but not communicating, just as a person speaking French cannot be understood by a person who speaks only Japanese.

A Real-World Example: Opening a Website

Let's connect the basic concepts to something we do every day. Suppose we open the given link on laptop:

www.example.com

At a very simplified level:

Your Laptop → Wi-Fi / Ethernet → Router → Internet → Web Server → Website Data

The actual process is much more complicated, but this simple model gives us the basic idea.

Basic Networking Terminology

Networking has its own vocabulary. Understanding these terms is essential before moving to configuration and troubleshooting.

1. Node

A **node** is a device or connection point that participates in a network. Examples include:

- Computer
- Laptop
- Server
- Router
- Switch
- Printer

Consider: **PC1 → Switch → PC2**

The devices participating in this network can be considered network nodes.

Remember

> **Node = a device or point participating in a network.**

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2. Host

A **host** is a network-connected device that can send or receive data over a network. Examples:

- Desktop computer
- Laptop
- Smartphone
- Server
- Networked printer

For example, if a laptop is connected to a university network and communicates with a server, the laptop is a **host**.

Node vs. Host:

These terms are closely related. A useful beginner-level distinction is:

- **Node** is a broad term for a participating network device/point.
- While **host** generally refers to an end device that communicates on the network.

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3. Client

A **client** is a device or application that **requests a service or resource** from another device or application. For example, when you open a website:

Web Browser → Requests webpage → Web Server

Your browser/device acts as the **client**.

Examples

- Web browser requesting a webpage
- Email application requesting email
- Student computer requesting a file from a server

---

4. Server

A **server** is a computer or software system that provides services or resources to other devices, called clients. Examples:

- Web server
- File server
- Database server
- DNS server
- Email server

When you visit a website:

Client → Request → Server

Server → Response → Client

Client vs Server:

Client	Server
Requests a service	Provides a service
Usually initiates a request	Usually responds to requests
Example: Web browser	Example: Web server
Example: Student PC	Example: University server

Easy analogy:

Think about a restaurant

Customer → requests food → Restaurant

Customer = **Client**

Restaurant = **Server**

The client requests something, while the server provides it.

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5. Packet

A **packet** is a small unit of data transmitted across a network. Large amounts of information are generally divided into smaller pieces so that they can be transmitted through the network. For example:

Large File

↓ (divided into)

Packet 1 | Packet 2 | Packet 3 | Packet 4 | ...

These pieces travel through the network toward their destination. Network data is divided into manageable units for transmission.

Important

Do not think of a packet as simply "a file." A packet is a unit of network communication containing part of the information being transmitted along with information needed for delivery.

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6. Protocol

A **protocol** is a set of rules that defines how devices communicate. Just as humans need agreed rules for communication, computers need communication protocols. For example, two people communicating might agree that:

1. One person speaks.
2. The other listens.
3. The listener responds.
4. Both understand the language being used.

Networking protocols perform a similar role for computers. They define things such as:

- How data is formatted
- How devices communicate
- How information is exchanged
- How communication is controlled
- How errors may be handled

Examples

- HTTP
- HTTPS
- TCP
- IP
- DNS
- DHCP

Remember

> **Protocol = rules for communication.**

---

7. Bandwidth

Bandwidth refers to the maximum amount of data that a network connection can theoretically carry per unit of time. It is commonly expressed in:

- bits per second (bps)
- Kbps
- Mbps
- Gbps

Suppose an Internet connection is advertised as **100 Mbps.** This indicates a theoretical maximum capacity of approximately 100 million bits per second under the relevant conditions.

Analogy:

Imagine a highway.

A highway with many lanes can carry more vehicles at the same time.

Similarly:

> **Higher bandwidth → greater potential data-carrying capacity.**

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8. Latency

Latency is the time it takes for data to travel from one point to another. It is commonly measured in:

> **milliseconds (ms)**

Suppose a network request takes approximately **20 ms** to travel from one device to another. That represents the network's latency for that communication measurement.

Analogy: Think of two cities.

Even if a highway can carry thousands of cars, a car still takes some amount of time to travel between the cities. That travel time is analogous to **latency**.

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9. Throughput

Throughput is the actual amount of data successfully transferred through a network over a given period. This is different from bandwidth.

Suppose a connection has **100 Mbps bandwidth** but due to network conditions, congestion, protocol overhead, or other limitations, only **70 Mbps** of useful data is actually transferred. The actual transfer rate is the **throughput**.

Bandwidth vs Latency vs Throughput

Concept	Meaning	Example
Bandwidth	Maximum potential capacity	100 Mbps
Latency	Delay in communication	20 ms
Throughput	Actual successful data transfer	70 Mbps

Easy Analogy:
Think of a highway:

- **Bandwidth** = capacity/number of lanes
- **Latency** = travel time
- **Throughput** = actual traffic successfully moving

Remember

- Bandwidth tells us how much the connection can potentially carry.
- Latency tells us how long communication takes.
- Throughput tells us how much data is actually transferred.

Wired and Wireless Networks

Networks can broadly be categorized based on how devices connect.

1. Wired Network

A **wired network** uses a physical cable to connect devices. Common technologies include Ethernet.

Example: **PC → Ethernet Cable → Switch**

Advantages

- Stable connection
- Generally reliable
- Less affected by wireless interference
- Suitable for fixed devices

Examples

- University computer laboratory
- Office network
- Data center

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2. Wireless Network

A **wireless network** allows devices to communicate without a physical network cable between the device and wireless access point. Wi-Fi is a common example.

Example: **Laptop)))) Wi-Fi)))) Access Point**

Advantages

- Mobility
- Easier connection for mobile devices
- Less cabling
- Convenient for users

Examples

- Home Wi-Fi
- University Wi-Fi
- Public Wi-Fi

Wired vs. Wireless

Wired	Wireless
Uses physical cables	Uses wireless signals
Usually more stable	Provides mobility
Requires cabling	Less physical cabling
Ethernet is common	Wi-Fi is common

Network Types Based on Geographic Area

Networks can also be classified according to their geographical coverage.

1. LAN — Local Area Network

A **LAN** is a network covering a relatively small geographical area.
Examples:

- Home
- Computer laboratory
- Office
- Single building

For example: **PC → Switch → PC → Printer**

This could represent a small LAN.

Key idea

LAN = Local / small area

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2. MAN — Metropolitan Area Network

A **MAN** covers a larger area than a typical LAN and is associated with a metropolitan or city-wide area.
Example:

- A network connecting multiple buildings or facilities across a city.

Key idea

> **MAN = Metropolitan / city-scale area**

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3. WAN — Wide Area Network

A **WAN** covers a large geographical area. It can connect:

- Cities
- Countries
- Continents

The Internet is the most prominent example of a very large interconnected network.

Key idea

> **WAN = Wide geographical area**

LAN vs MAN vs WAN

Feature	LAN	MAN	WAN
Coverage	Small	Metropolitan/city	Large geographical area
Example	Office	City-wide network	Inter-city/country network
Typical purpose	Local connectivity	Connect sites across metropolitan area	Connect geographically distant networks

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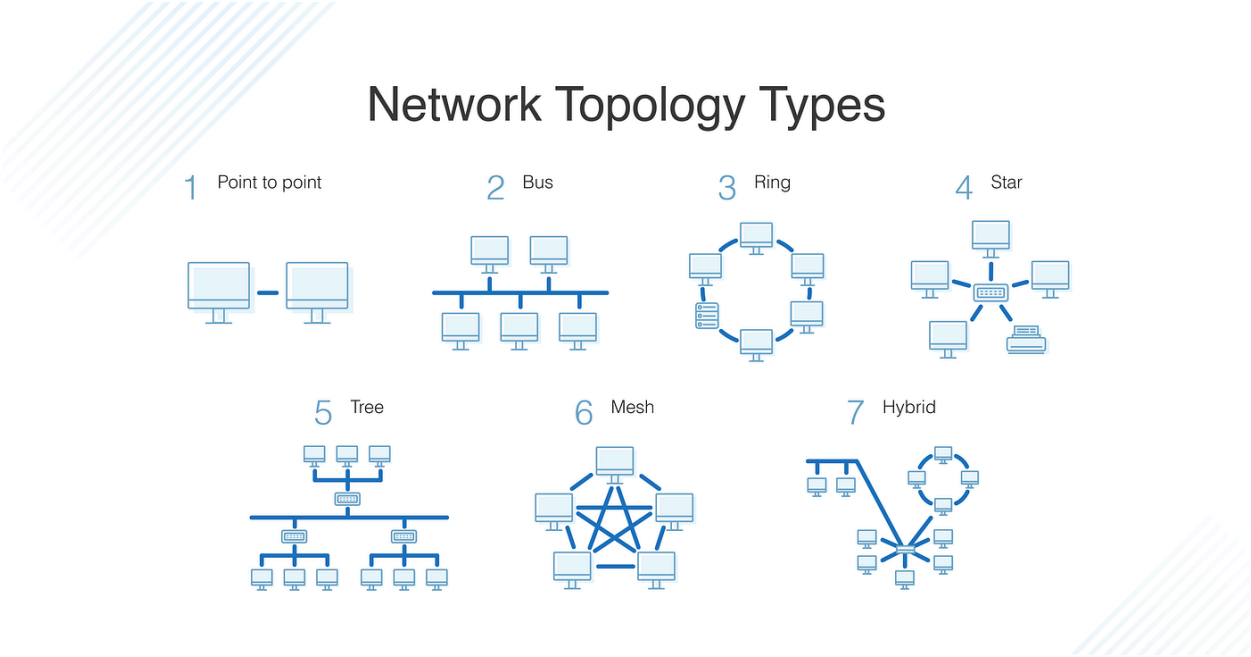
Network Types Based on Geographic Area

The Topology is the geometric representation of the relationship of the links and linking devices.

OR

Topology defines physical or logical arrangement of links in a Network.

- The ways in which the wires can be run in the network to link with the computer.
- The ways in which the connections are made in the network.
- Simply the physical layout of the network.



Internet

The **Internet** is a global system of interconnected networks that communicate using standardized networking technologies and protocols. It connects enormous numbers of:

- Computers
- Servers
- Networks

- Smartphones
- IoT devices
- Data centers

Through the Internet, users can access services such as:

- Websites
- Email
- Cloud applications
- Streaming
- Online learning
- Social media

Important idea

- The Internet is not one single physical network.
- It is a **network of interconnected networks**.

Intranet

An **Intranet** is a private network used within an organization. For example, a university might have an internal network containing:

- Internal portals
- Staff systems
- Internal databases
- File-sharing systems

These services may not be publicly accessible from the Internet.

Internet vs Intranet

Internet	Intranet
Public/global network environment	Private organizational network
Used by users worldwide	Used by authorized organization members
Examples: public websites	University/company internal systems

Easy way to remember

- Internet = public/global
- Intranet = private/internal

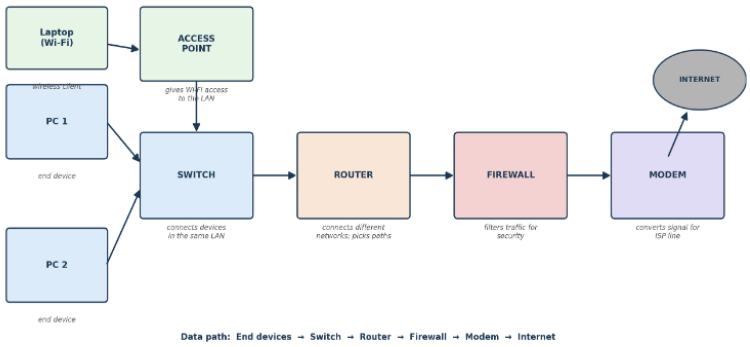
Network Components and Devices

A computer network requires different hardware components to allow devices to connect and communicate.

For example:

- NIC
- Hub
- Switch
- Router
- Modem
- Access Point
- Firewall
- Gateway

These devices will become increasingly important as we build networks.



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1. NIC — Network Interface Card

A **Network Interface Card (NIC)** is the hardware interface that allows a device to connect to a network. A computer may have:

- Ethernet NIC
- Wireless NIC

For example:

Laptop → Wi-Fi NIC → Wireless Network

or

Desktop PC → Ethernet NIC → Ethernet Cable → Switch

Key idea

NIC = provides the network connection/interface for a device.

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2. Hub

A **hub** is a basic networking device used to connect multiple devices. When a hub receives data, it sends the data out through its ports rather than intelligently selecting only the intended destination port.

Example: **PC1 → Hub → PC2, PC3, PC4**

The connected devices may all receive the transmitted signal.

Important

Hubs are largely obsolete in modern Ethernet networks, but understanding them helps students understand why switches are more efficient.

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3. Switch

A **switch** connects devices within a local network and forwards Ethernet frames based on MAC addresses.

Example: **PC1 → Switch → PC2**

The switch learns which MAC addresses are associated with its ports and uses this information to make forwarding decisions.

Key idea

Switch = connects devices within a LAN and intelligently forwards Ethernet frames.

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4. Router

A **router** connects different networks and forwards packets between them.

Example:

LAN 1 → Router → LAN 2

or:

Home Network → Router → Internet

A router makes forwarding decisions based on network-layer information, particularly IP addressing.

Remember

- Switch → connects devices/networks within a local environment
- Router → connects different IP networks

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5. Modem

The word **modem** comes from **Modulator + Demodulator.** A modem is used to provide communication between a local network/device environment and certain service-provider communication systems.

A simplified home setup might look like:

Home Devices → Router/Network → Modem → ISP

The exact equipment and architecture can vary depending on the Internet service.

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6. Access Point

An **Access Point (AP)** provides wireless connectivity to devices. Example:

Laptop)))) Smartphone)))) → Access Point → Wired Network

The access point allows wireless devices to join the network.

Key idea

> **Access Point = provides wireless network access.**

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7. Firewall

A **firewall** is a security mechanism that controls network traffic according to defined rules. It can allow or block traffic based on factors such as:

- Source
- Destination
- Protocol
- Port

For example, a firewall may allow legitimate web traffic while blocking unauthorized traffic.

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8. Gateway

A **gateway** is a device or system that provides a path between different networks or networking environments. A gateway can act as an entry/exit point from one network toward another. A common example is the **default gateway**:

PC → Default Gateway → Other Network

MAC Address and IP Address

Network devices need ways to identify themselves and communicate with other devices. Every device on a network needs two kinds of “address” to be found:

1. MAC Address
2. IP Address

1. MAC Address

A MAC (Media Access Control) address is an identifier associated with a network interface. A MAC address is commonly represented in hexadecimal notation.

Example: 00:1A:2B:3C:4D:5E

MAC addresses are important in local Ethernet communication. Switches use MAC addresses to learn and forward Ethernet frames.

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2. IP Address

An **IP address** is a logical address used to identify a device/interface within an IP network.

Example IPv4 address: 192.168.1.10

IP addressing allows devices to communicate within networks and across interconnected networks.

Important:

- **MAC address** → associated with network interface
- **IP address** → logical network address

	IP Address	MAC Address
Assigned by	Software (network/administrator, static or via DHCP)	Hardware manufacturer (burned into the NIC)
Can it change?	Yes, it can be reassigned.	No, permanent
Format example	192.168.1.10	AA:BB:CC:11:22:33
Used for	Finding a device across different networks (like a postal address).	Finding a device within the same local network (like a fingerprint). It is the identity of the network interface

	It tells where the device is on the network	

Network Ports

A **network port** helps identify a particular service or application on a device.

Think of it this way:

- IP address → identifies the network destination/device
- Port → identifies the service/application

Examples

Service	Common Port
HTTP	80
HTTPS	443
DNS	53

For example, a server may simultaneously provide multiple services. The port number helps networking software determine which service should handle incoming traffic.

Basic Networking Protocols

A protocol defines rules for communication. Some important protocols are:

- **HTTP**
  - Used for web communication.
- **HTTPS**
  - Used for secure web communication.
- **DNS**
  - Used to translate domain names into IP addresses.
  - Example:

    - **www.example.com → IP address**
- **DHCP**
    - Used to automatically provide network configuration information to devices.
- **TCP**
    - Provides reliable transport-level communication.
- **IP**
    - Provides logical addressing and routing of packets between networks.
- **Ethernet**
    - A major technology used for communication within wired LANs.

Why Do We Need Networking Models?

Computer networking involves many technologies and processes. For example, when a laptop communicates with a web server, several things happen:

- Application creates data
- Data is prepared for transmission
- Transport communication is handled
- IP addressing is used
- Ethernet/Wi-Fi carries the data
- Physical signals transmit the information

Trying to understand everything at once would be difficult. Therefore, networking is organized into **layers**.

Benefits of layering

Layering helps us:

- Divide a complex process into manageable parts
- Understand the responsibility of each layer
- Design networking technologies
- Troubleshoot problems
- Standardize communication

OSI Model

The **OSI (Open Systems Interconnection) Reference Model** is a conceptual model that divides network communication into **seven layers**.



Layer 7 — Application

Provides network services that applications use to communicate. Examples include services associated with:

- Web communication
- Email
- File transfer

Simple idea

> Application = network services used by applications

Layer 6 — Presentation

Concerned with how data is represented. It can involve concepts such as:

- Data formatting
- Encoding

- Encryption
- Compression

Simple idea

> Presentation = how data is represented

Layer 5 — Session

Concerned with establishing, managing, and ending communication sessions.

Simple idea

> Session = manages communication sessions

Layer 4 — Transport

Responsible for end-to-end transport of data between applications. Important concepts include:

- Reliability
- Segmentation
- Flow control

TCP and UDP operate at the transport layer.

Simple idea

> Transport = end-to-end delivery

Layer 3 — Network

Responsible for logical addressing and routing. IP operates at this layer.

Simple idea

> Network = addressing and routing

Layer 2 — Data Link

Responsible for communication over a local network link. Important concepts include:

- Frames
- MAC addresses
- Ethernet

Simple idea

> Data Link = local network delivery

Layer 1 — Physical

Concerned with the actual transmission of bits through a physical or wireless medium. Examples include:

- Cables
- Connectors
- Electrical signals
- Radio signals

Simple idea

> Physical = transmission of bits/signals

Easy Way to Understand the OSI Model

Instead of memorizing seven definitions, think about sending a package.

Application: What do I want to send?

Presentation: What format should it use?

Session: How do we establish/manage the communication?

Transport: How should the data be delivered end-to-end?

Network: Where does it need to go?

Data Link: How do I deliver it across this local link?

Physical: How do I physically transmit it?

The analogy is not a perfect technical mapping, but it will help you understand why communication is divided into layers.

Example:

1. **Application Layer** (Layer 7)
   This is where the data starts. You're typing a WhatsApp message or loading a webpage. The application (WhatsApp, Chrome) prepares the actual content you want to send.

2. **Presentation Layer** (Layer 6)
   The data gets formatted, encrypted, or compressed so it's in a format the receiving system can understand, like translating your message into a universal "language" and locking it if needed (encryption).

3. **Session Layer** (Layer 5)
   A "session" or conversation is established between your device and the receiver's device, think of it like dialing a call and making sure the line stays open until the conversation ends.

4. **Transport Layer** (Layer 4)
   The data gets broken into smaller pieces called **segments**, and each one is tagged with information ensuring reliable delivery (this is where protocols like TCP make sure nothing is lost or arrives out of order).

5. **Network Layer** (Layer 3)
   Each segment gets a **destination address (IP address)** like writing the destination address on an envelope so it knows which network/device to travel to.

6. **Data Link Layer** (Layer 2)
   The data is packaged into **frames** and given a local address (MAC address) like writing which specific mailbox on the street it needs to go to.

7. **Physical Layer** (Layer 1)
   The data is finally converted into raw **electrical signals, light pulses, or radio waves** and physically transmitted through a cable, fiber optic line, or Wi-Fi signal.

TCP/IP Model

The **TCP/IP model** is associated with the protocol architecture used by the Internet and modern computer networks. A commonly taught four-layer version is:

1. Application Layer

Provides network services to applications. Examples include:

- HTTP/HTTPS
- DNS
- DHCP

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2. Transport Layer

Provides communication between applications on different devices. Important protocols:

- TCP
- UDP

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3. Internet Layer

Responsible for:

- IP addressing
- Packet forwarding
- Routing between networks

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4. Network Access Layer

Concerned with local network communication and physical transmission. Examples include:

- Ethernet
- Wi-Fi

OSI vs. TCP/IP

